Activity 1A: Blinking an LED

Purpose:

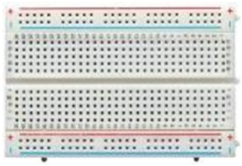
- To wire and code a light to blink at different intervals using parallel and series circuits.
- To use BlocklyDuino and Codebender to create code to communicate with the Arduino microcontroller

Materials: Computer, Arduino Uno Board, USB cable, Breadboard, 2 coloured LED's, 200 - 330 Ω Resistor, 2 Jumper Wires

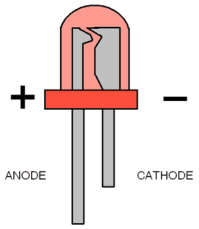
New Components:



Arduino Uno Board: The Arduino is a microcontroller that will communicate the code to the components of your program. For example, the uno will communicate your code to the LED so that it performs what it is instructed to.



Breadboard: A breadboard is a solderless device for temporary wiring used to test circuit designs. Most electronic components in electronic circuits can be interconnected by inserting their leads or terminals into the holes and then making connections through wires where appropriate.



LEDs: (Light Emitting Diodes) are small lights made from a silicon diode. They come in different colors, brightnesses and sizes. LEDs have a positive (+) leg and a negative (-) leg, and they will only let electricity flow through them in one direction. The (+) leg is the longer of the legs on the LED.

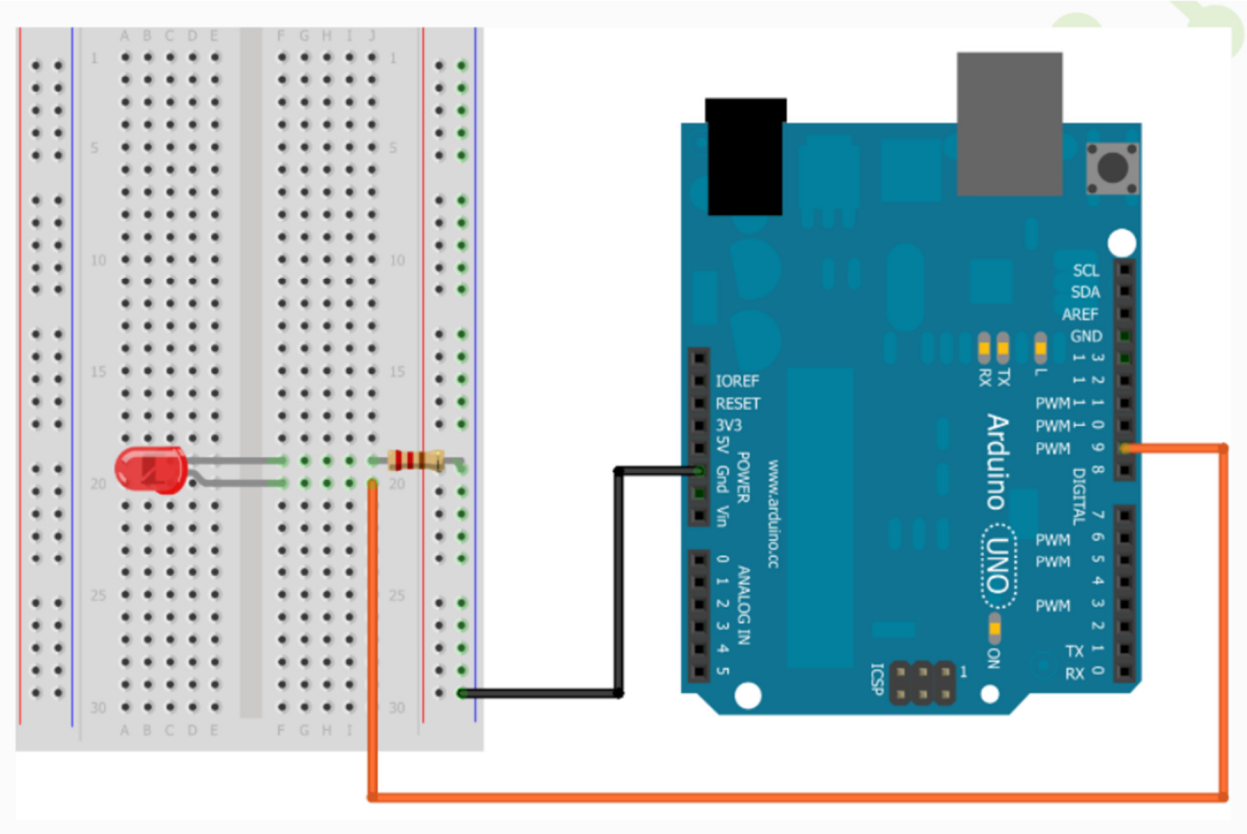
Resistors: Resistors resist the flow of electricity. You can use them to protect sensitive components like LEDs from . The strength of a resistor (measured in ohms) is marked on the body of the resistor using small colored bands. Each color stands for a number, which you can look up using a resistor chart.



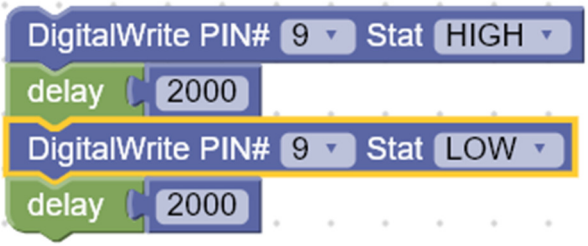
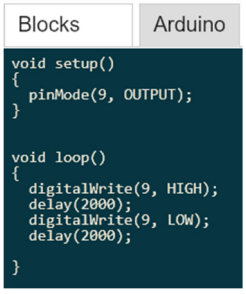
Assignment 1A: BLINKY

Procedure:

1. Wire the project using the following schematic.



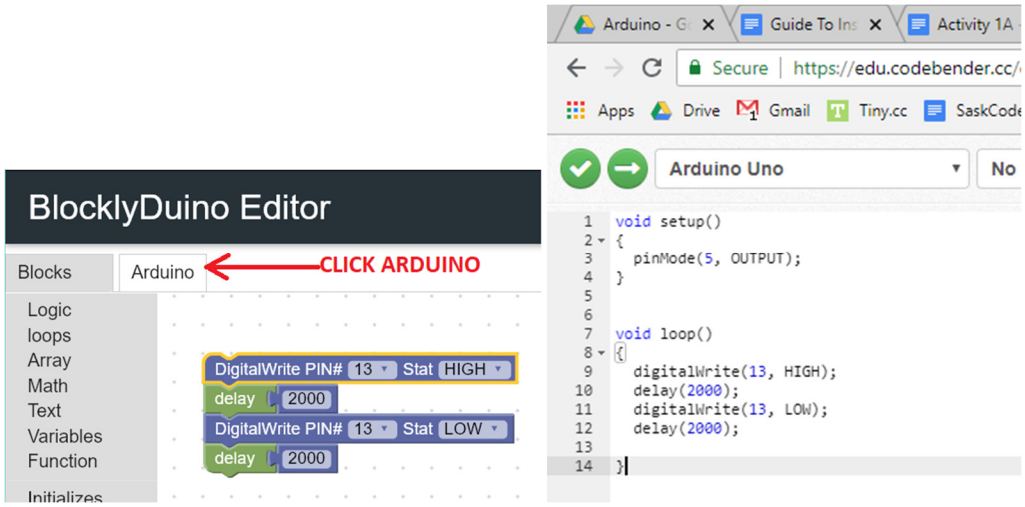
2. You can create your code in BlocklyDuino that you downloaded last lesson and then paste it to the Codebender website. Below is the blockly code that you will need to make and the Arduino code that you will need to copy and paste into codebender. Click BlocklyDuino Reference for a link to all the blockly codes. You can also watch the following video to help you with the process:
   VIDEO: Blinky

BLOCKLY CODE	ARDUINO CODE
	

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*****SAVE your sketch to your BlocklyDuino Folder that you created*****

3. Copy the Arduino Code from BlocklyDuino into the Codebender Website. To paste the code into the website be sure to use ctrl "v." Hook up your arduino to your computer using the USB cable.



6. Upload your program to the arduino. Make sure the Arduino Uno board is hooked up using the supplied USB cord to your computer and click the green "upload" button.

7. Your program is now loaded. Observe your LED and complete your worksheet titled Assignment 1A: Blinking an LED.

Troubleshooting: If your circuit is not working!

1. Double check your code! Did you put every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit! Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
4. Next check each component. Check the led, is the long positive arm connected to pin 13?
5. Next inserting the LED directly into pin 13 and the adjacent ground. Does it light when the code is on?
6. Next check the individual wires in the circuit. This can be done by connecting the LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.
7. For further information on using the serial monitor see Activity 1A (Part 2) Using a Serial

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Monitor Display.

Assignment 1A WORKSHEET: Blinking an LED.

1. Describe in detail what the LED is doing. (Observe the timing).

2. THE CODE. Code is a set of instructions used to communicate to the arduino. Answer the following questions relating to code:

   a) What does the output block 

do?

   b) How would you write this code in Arduino?

   c) What does this timing block 

do ?

   d) How would you write this code in Arduino?

   e) The following code turns the led off 

. How would you write this in Arduino. (two lines)

   f) Coders can add a description of each line of code by putting // after their code. It gives the person a reading it a description of what the code does. What punctuation does Arduino use at the end of each instruction to separate them?

Extension Challenges:

3. Change your code so that your LED flashes on and off at one second intervals. Explain how you changed your code to achieve this.

4. Change your code so that your LED flashes on for three seconds and off for one second. Explain how you changed your code to achieve this.

5. Connect a different colored light to pin 10. Code your arduino to flash both lights at the same time.

6. Code light one to flash every two seconds and light two to code every second.

7. Code your lights so that when light one flashes on, light two flashes off and when light two flashes on, light one flashes off.

CONCLUSION:

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8. List two real life applications that coding lights have in our everyday lives.

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